

NAYSAN MUNJE

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SUMMARY OF QUALIFICATIONS

- **Autonomous Vehicles** Research Assistant with over a year of experience developing multi-sensor data collection systems and facial recognition software.
- Three years of experience with **SolidWorks**, including advanced surfacing, motion studies, 3D printing, and CNC machining.
- Designed, soldered, and programmed custom PCB projects using **KiCad**, **Arduino** (UNO, ESP32, ATTiny), and **Raspberry Pi**.
- Proficient in **Python**, **MATLAB**, and **C** for mechanical control, automation, and data analysis.
- Built complex robotic mechanisms, including multi-stage elevator systems and articulated robotic arms, for competitive applications.
- Familiar with a variety of communication protocols including **I2C**, **UART**, **SPI**, and **CAN**.

EXPERIENCE

NSERC Research Internship, Autonomous Vehicles

May 2025 – Aug 2025

Advanced Biomechatronics and Locomotion Lab, Carleton University

Ottawa, ON

- Engineered a vehicle-integrated **PCB data logger** for real-time high speed data capture, including GPS, IMU, and CAN bus signals for machine learning models with **ESP32** and **Raspberry Pi** infrastructure used in published **PhD Thesis**.
- Collected, processed and filtered video and sensor data with **MATLAB** and **Python** for use in **academic papers**.
- Designed and assembled multiple prototype PCBs, applying precision soldering techniques used for rapid testing.

Undergraduate Research Assistant, Autonomous Vehicles

Aug 2024 – Present

Advanced Biomechatronics and Locomotion Lab, Carleton University

Ottawa, ON

- Created AI-based face detection and automated driving algorithms in **MATLAB**.
- Integrated computer vision systems into ongoing autonomous driving research projects.

Mechanical Design Lead / Mentor

May 2022 – Feb 2025

FIRST Robotics Competition

Ottawa, ON

- Led a **20-member** student team in designing, prototyping, and manufacturing competition robots with **SolidWorks**.
- Developed large-scale mechanisms, including a chain-driven three-stage elevator and high-precision robotic arms.
- Represented the team as **Robot Operator**, achieving a provincial finalist position and qualifying for the **World Championship** in Houston, TX.

PROJECTS

Car Data Sniffer

May 2025

ESP32, MATLAB, Python

- Built Bluetooth-enabled data logger capturing GPS, IMU, CAN bus, and camera data for AI model training.

Dice Roller

Oct 2024

Arduino, KiCad, SolidWorks

- Developed ATTiny powered custom PCB with LED display in a 3D-printed enclosure.

Elevator Robot

Feb 2023

SolidWorks

- Designed a 5-foot chain driven elevator mechanism enabling multi level object placement in competition robots.

EDUCATION

Bachelor of Mechanical Engineering

2024 - 2028

Carleton University - 3.9/4 GPA

Ottawa, ON

SHAD Canada

Jul 2023

National STEM enrichment summer program for grade 10 and 11 students

Toronto, ON